DhakaSim: heuristic evaluation against Nielsen's ten usability heuristics

An inspection of the DhakaSim Python build as it stands on 15 October 2026: its start screen, run screen, map-import dialog, help page and run report, judged heuristic by heuristic, with every open issue rated on Nielsen's severity scale and a short list of what to fix first. ASM Nuruzzaman.

Method

The evaluation follows Nielsen's heuristic evaluation procedure: one evaluator walks the interface twice, first to learn its flow and then to inspect each element against the ten heuristics, recording every violation with the heuristic it breaks and a severity. The owner set these heuristics as the basis for judging the screens on 6 October 2026, and the build was reworked against them over the following nine days; this document records where it has reached.

The walk covered six tasks: choose a junction and set up a run; change settings and take changes back; start, pause, zoom and switch views in a run; load the junction up while it plays; import a junction from a map; and find help. Three screens are reproduced below as they looked on the evaluation laptop (1536 by 793 pixels).

Severity	Meaning, after Nielsen
0	Not a usability problem.
1	Cosmetic. Fix only if there is time.
2	Minor. Low priority.
3	Major. Important to fix.
4	Catastrophic. Must fix before release.

One evaluator finds about a third of the problems a panel of five would, by Nielsen's own figures. Treat the issue list as a floor, not a census. No user study has been run.

The screens inspected

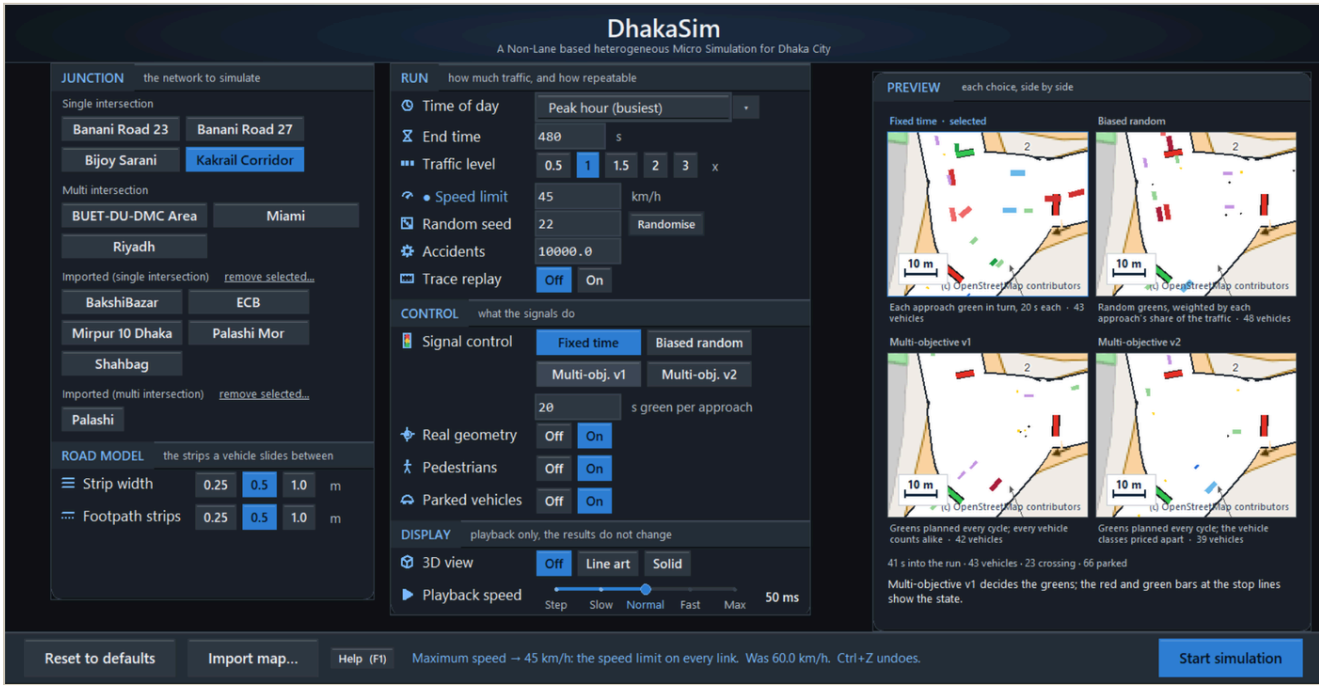


Figure 1. The start screen after the speed limit was changed to 45 km/h: the row carries a dot, the footer says what moved, what it was and that Ctrl+Z undoes, and the pictures beside the settings show the four signal controllers for the row under the pointer.

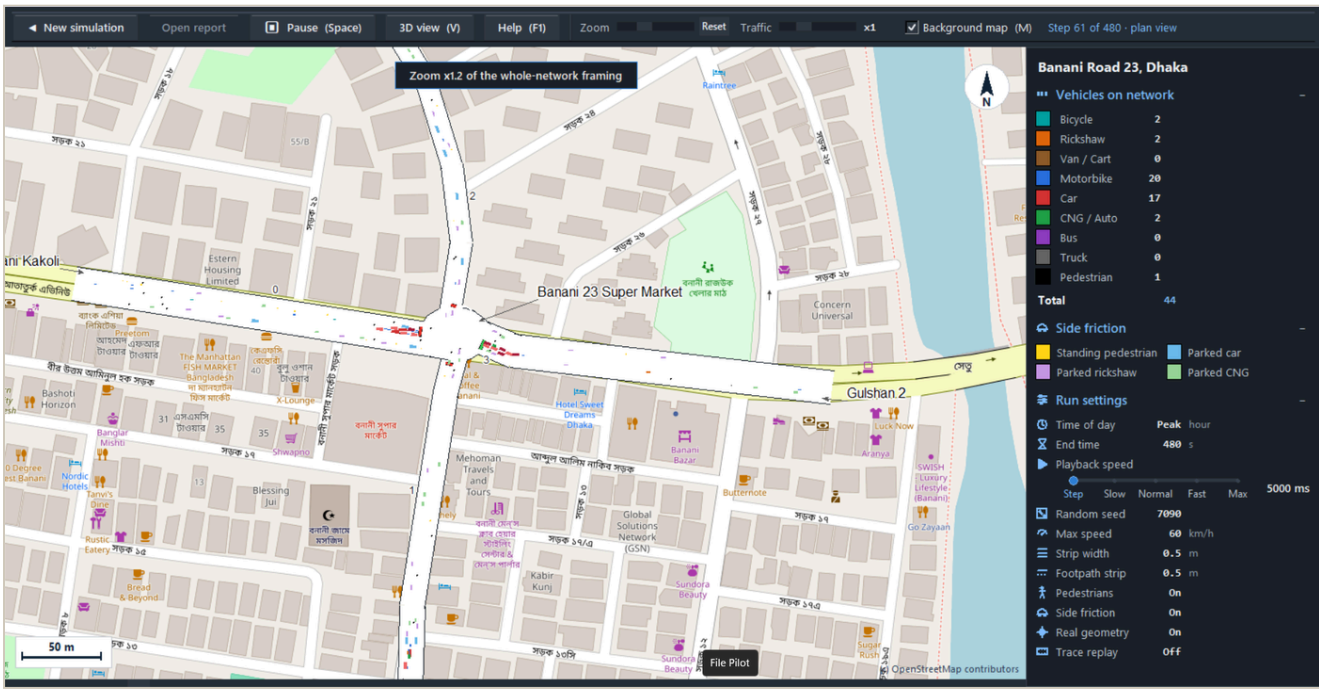


Figure 2. The run screen: the toolbar with the view, help, zoom, traffic and map controls and the step counter; the plan view over the map with a toast; the sidebar with the live counts, the side-friction key and the run's settings.

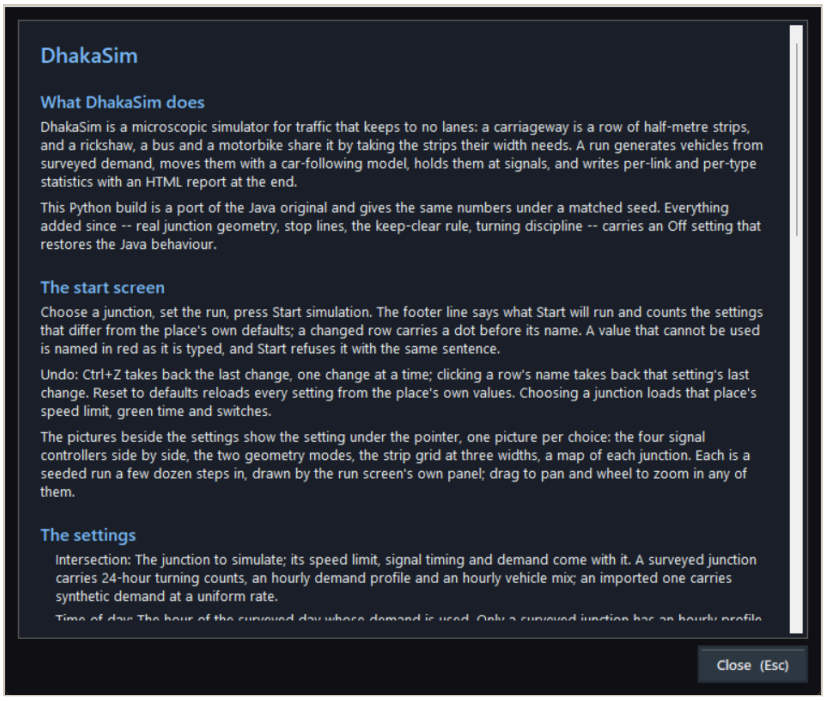


Figure 3. The help page, opened with F1 from either screen.

The ten heuristics

1. Visibility of system status

Rating: strong.

The start screen's footer always says what Start will run, counts the settings that differ from the place's defaults, and marks each changed row with a dot. Every change produces a sentence naming the setting, its new value, its meaning and its old value. On the run screen the toolbar shows the step, the end time and the view every simulation step; a toast on the picture reports every zoom, view change, map toggle, traffic change and undo; the bottom bar shows progress. The import dialog has a progress bar driven by the worker's own log lines, and the map-fetch button changes its caption while it works.

- 1 A preview cell says "Building..." while a second run is built but gives no sense of how long; on the largest network that is about a second.
- 1 The multi-objective scheduler re-plans every cycle at every junction, and nothing on screen shows a plan being made or what it chose.
- 1 On a window under about 1400 pixels wide the step counter is clipped at its end; the step itself stays visible.

2. Match between system and the real world

Rating: adequate.

Settings are worded as a reader thinks about them: playback speed as Step, Slow, Normal, Fast and Max rather than a frame delay in milliseconds; real geometry explained as medians and a roundabout ring rather than "Java parity"; places by name, each with a sentence about its surroundings; speeds in km/h, greens in seconds, traffic as a multiple of the surveyed rate. The signal controllers are named by what they do in the preview captions.

- 0 Closed the same day: the two cells now read "Planned, all alike" and "Planned, by class"; the paper's v1 and v2 stay in the help and the report.
- 1 "Strip width" is the model's own vocabulary. The picture beside it shows the grid, but the name does not say that a strip wider than any vehicle is a lane.
- 1 "Accidents" is set as close encounters per accident, so a larger number means fewer accidents. The caption says so; the direction is still the reverse of what the name suggests.

3. User control and freedom

Rating: adequate.

Ctrl+Z takes back the last change on the start screen, one change at a time, and the footer names what came back; clicking a row's name takes back that row's last change, and the name shows a hand while there is one. A typed value is one change however many keystrokes it took, and choosing a different junction is itself undoable. Reset to defaults reloads the place's own values. New simulation leaves a run at any point, asking first if it is unfinished; a view change reverses with the same key; the import picker has Close and Rebuild; imported networks can be removed and shipped ones cannot.

- 0 Closed the same day: Reset to defaults now asks when anything differs from the place's defaults, naming how many settings and which.
- 1 There is no redo. An undo too many has to be re-typed.
- 1 Choosing a junction drops the undo history of the settings that junction's defaults rewrote (speed limit, green time, the two switches). This is deliberate, since the old values belonged to the other place, but nothing says so.
- 1 The run screen's live controls (zoom, traffic, playback speed) have no undo; Reset covers the zoom only.

4. Consistency and standards

Rating: adequate.

One palette and one set of controls across the start screen, the run screen, the import dialog, the help page and the HTML report. The same five-detent slider sets playback speed on both screens; the same row icons appear in the form and the sidebar; every choice is a row of cells with one filled and every switch an Off/On pair; keys are named on their buttons; Esc closes the help as it closes the import dialog.

- 0 Closed the same day: the sidebar's Run settings use the form's names, Speed limit, Parked vehicles and Footpath strips.
- 1 The zoom and traffic sliders are standard Tk scales and the playback slider is a drawn one; they look alike but the drawn one moves in detents and answers the arrow keys, the scales move continuously.

5. Error prevention

Rating: strong.

A value that cannot be used is named in red as it is typed, and Start refuses it with the same sentence, so a traceback never reaches the reader. The junction picker turns its status red when a chosen junction has fewer than three legs and refuses a set of junctions that is not connected before anything is fetched. The map fetch refuses to overwrite hand-placed imagery. A playback value the slider does not offer is kept

rather than snapped. Nothing the preview runs can leak into the run: a seeded run after previewing is byte for byte the run a fresh screen gives, and a test pins it.

- 1 Trace replay can be switched on with a recording of another place; the preview explains it instead of preventing it, and the run then shows an empty road.
- 1 A seed of -1 is accepted and draws a seed that is never shown, so that run cannot be repeated; the caption says so.

6. Recognition rather than recall

Rating: strong.

Every choice of every setting has a picture beside it: the four signal controllers side by side with a line each, the two geometry modes, the strip grid at three widths, the map of each junction with a sentence on its surroundings, the five playback speeds with their milliseconds. The footer shows a row's explanation while the pointer is on it. The sidebar keeps the run's settings in view for the whole run, with the same icons as the form, and the report repeats them.

- 1 The pictures need width. At 1280 pixels the form scrolls beside a narrower column and the pictures shrink to a size where the strip grid is hard to read.
- 1 On a 720p window the sidebar drops to 8 point type to stay in view, and on anything shorter it scrolls.

7. Flexibility and efficiency of use

Rating: adequate.

Keyboard shortcuts for pause, view, map, help and undo; wheel zoom and drag pan; Return in the import dialog's Where box previews the place. For anyone who does not need the window, a per-network defaults file, a command-line override for any parameter, a headless mode and a sweep harness. Randomise draws a fresh seed in one click.

- 1 No saved presets: a form set up for a particular study has to be set up again next time, or written into the network's defaults file by hand.
- 1 No recent-places list in the import dialog; a place imported last week is typed again.

8. Aesthetic and minimalist design

Rating: strong.

The form chooses its density by measurement, so nothing is clipped and nothing is wasted; captions move to the footer when they do not fit; the Scale row left the form because it meant nothing to a reader, and the sentence under the map toggle went for the same reason; the picture column takes a third of the window and no more; the sidebar is built at the largest type that fits.

- 1 Thirteen junction tiles on this laptop push the form to a tight rung, with 2 pixels of width to spare.
- 0 On a large monitor every row carries a two-line caption; the help page now says the same, so the captions could be shorter there.

9. Help users recognise, diagnose and recover from errors

Rating: adequate.

Error sentences name the setting and the rule ("End time must be a whole number of 1 or more") in the same place the status is shown. The import dialog reports which stage failed and leaves the network runnable when only the map or the road fit failed. A trace recording of another place is explained in the preview rather than shown as an empty road. A failed map fetch says so in a toast and leaves the button ready to try again.

- `0` Closed the same day: when every Overpass server turns the request away the dialog says the servers are busy and to try again in a minute or with a smaller radius; the server's own text goes to the log.
- `1` A failed import's reason sits in the dialog's log pane; the status line says only that it failed.

10. Help and documentation

Rating: adequate.

F1, or a Help button on either screen, opens a help page in the window: what the simulator does, the start screen and its undo, every setting with its meaning and unit, the run screen's keys and controls, importing a map, and where results go. The settings section is generated from the same table the footer reads, and a test checks every label is on the page. The README documents every parameter and quirk, and each run's report explains every result and term.

- `1` The help page is text only. A picture of each screen with its parts named would answer most first questions faster.
- `1` There is no link from a row to its help entry; a reader opens the page and scrolls to the setting.

Summary

Heuristic	Rating	Open issues	Worst severity	Headline
1. Visibility of system status	strong	3	1	Status everywhere; the scheduler's planning is the one silent process.
2. Match with the real world	adequate	2	1	The controllers are named by what they do; strip width is still the model's word.
3. User control and freedom	adequate	3	1	Undo and a confirmed Reset arrived on 15 October; no redo yet.
4. Consistency and standards	strong	1	1	One vocabulary on both screens; two kinds of slider remain.
5. Error prevention	strong	2	1	Bad values are refused in a sentence, never a traceback.
6. Recognition rather than recall	strong	2	1	A picture for every choice; needs a wide window.
7. Flexibility and efficiency	adequate	2	1	Keys, wheel, headless and sweeps; no presets.
8. Aesthetic and minimalist design	strong	2	1	Density by measurement; the laptop rung is tight.
9. Error recognition and recovery	adequate	1	1	Clear sentences for settings and for a failed fetch alike.
10. Help and documentation	adequate	2	1	Help arrived on 15 October; it has no pictures yet.

Twenty-four issues were found, none above severity 2. The four minor ones were fixed the same day and are marked closed above; the twenty that remain are cosmetic. The build has no major or catastrophic usability problem that this inspection could find.

What to fix first

The four minor findings were fixed on the day of the evaluation: Reset to defaults asks first, the sidebar uses the form's names, the planned controllers are named by what they do, and a failed map fetch says what to do. Of the cosmetic findings, these are worth the next hour:

1. **Add pictures to the help page**, one per screen with its parts named, and a link from each row's footer hint to its entry. (Heuristic 10, severity 1.)
2. **Add redo** beside undo, and a sentence in the footer when a junction change clears part of the history. (Heuristic 3, severity 1.)

3. **Show the scheduler's decision** as a toast when a planned controller sets a new cycle, so the one silent process has a voice. (Heuristic 1, severity 1.)
4. **Move the two Tk scales to the drawn slider**, so zoom, traffic and playback behave alike under the keyboard. (Heuristic 4, severity 1.)

> **What this evaluation does not cover.** It is one evaluator's inspection, not a test with users, and it says nothing about whether a transport engineer reaches the right conclusions from the numbers the screens show. It covers the Python build's own screens; the HTML report was read but not inspected element by element.

Heuristic evaluation of the DhakaSim Python build, 15 October 2026, against Jakob Nielsen's ten usability heuristics (1994) and his 0 to 4 severity scale. Screenshots taken on the evaluation laptop on the same day.